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Chapter 0: Introduction

1. Difference between selenium 3 vs selenium 4

Feature / Difference	Selenium 3	Selenium 4
Driver Setup	Requires setProperty()	No need for setProperty()
Communication Protocol	JSON Wire Protocol	W3C Protocol
Explicit Wait Time Units	Only seconds	Seconds, minutes, hours, etc.
Relative Locators	Not available	Available
Opening New Window/Tab	newWindow() not available	newWindow() available
Minimize Browser Window	minimize() not available	minimize() available

2. Java project vs Maven Project

Aspect	Java Project	Maven Project
Dependency Management	Manual (add JARs to classpath)	Automatic (via pom.xml)
Build Automation	Manual (compile/run manually)	Automated via Maven lifecycle
Folder Structure	No fixed structure	Standard directory layout
External JAR Handling	Add manually	Add in pom.xml, Maven fetches
Best For	Simple or learning projects	Real-world/test automation projects

3. CRM introduction

- CRM (Customer Relationship Management) is a software system that helps companies manage interactions with customers.
- B2B application
 - They don't sell the product to customers
 - They build applications and other business people use that to run their internal business.
 - CRM application acts like a template that can be used by other business people to run their business
 - By referring CRM, TestYantra will build an application based on requirements given by Qspiders.

◆ Key Features of a CRM System:

- Contact Management – Stores customer data like name, email, phone, etc.
- Sales Tracking – Tracks leads, opportunities, and deals.
- Marketing Automation – Sends campaigns and tracks results.
- Customer Support – Manages service tickets and queries.
- Analytics & Reporting – Provides reports on sales, service, performance.

◆ Examples of CRM Tools:

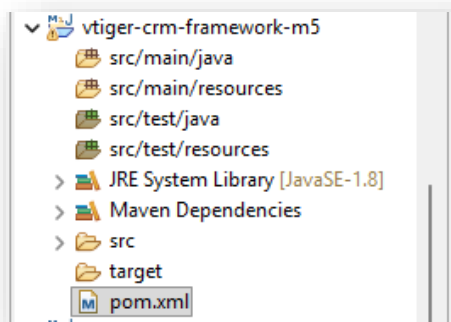
- a. Salesforce
- b. Vtiger CRM
- c. Zoho CRM
- d. HubSpot CRM
- e. Microsoft Dynamics 365

◆ Why CRM is Important:

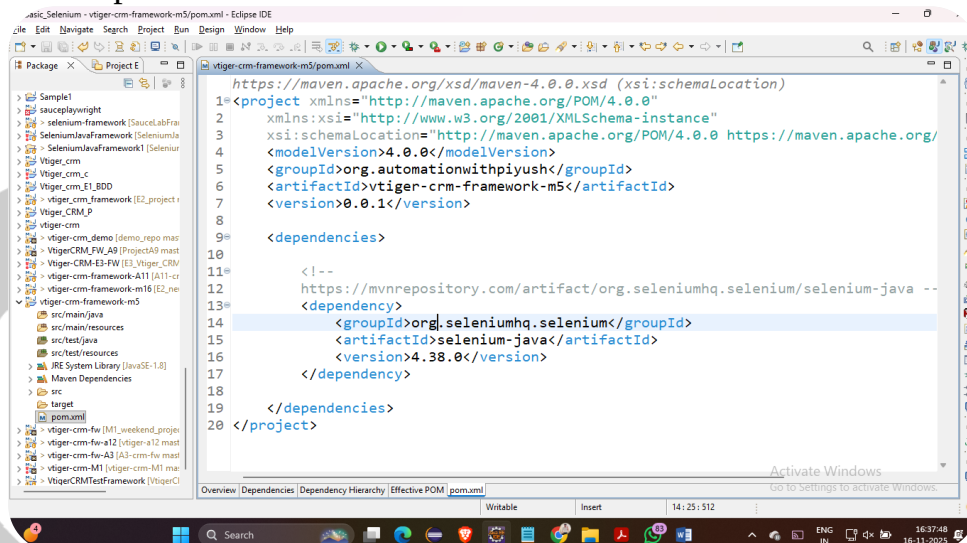
- Improves customer satisfaction
- Increases sales and productivity
- Helps in decision making using data
- Ensures better communication and follow-up

4. V-Tiger Test scripts

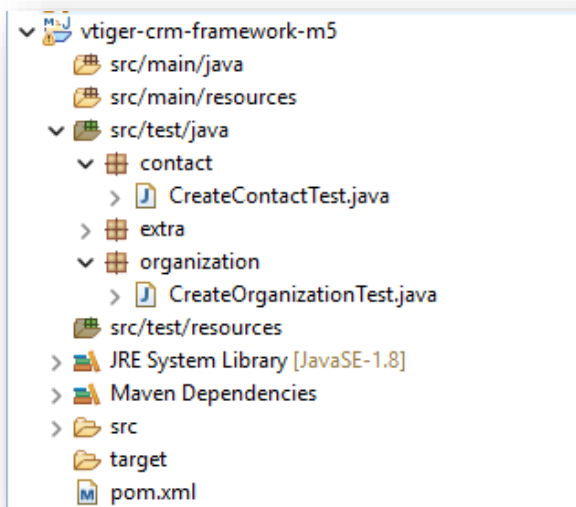
1. Maven project creation



2. Add dependencies



3. Create 2 classes inside 2 diff packages (contact & organization)



4. CreateContactTest.java

```
package contact;

import java.time.Duration;

import org.openqa.selenium.By;
import org.openqa.selenium.Keys;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.interactions.Actions;

public class CreateContactTest {
    public static void main(String[] args) throws InterruptedException {
        //      open the browser
        WebDriver driver = new ChromeDriver();
        driver.manage().window().maximize();
        driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(15));

        //      Login
        driver.get("http://localhost:8888/");

        WebElement username = driver.findElement(By.name("user_name"));
        WebElement password = driver.findElement(By.name("user_password"));
        username.sendKeys("admin");
        password.sendKeys("password" + Keys.ENTER);

        //      Create contact
        driver.findElement(By.linkText("Contacts")).click();
        driver.findElement(By.cssSelector("img[alt='Create Contact...']")).click();

        //      Filling Form
        String lastName = "Kumar";
        WebElement lastNameField = driver.findElement(By.name("lastname"));
        lastNameField.sendKeys(lastName);

        //      Save
        driver.findElement(By.cssSelector("input[title='Save [Alt+S]']")).click();

        //      Verification
        String actLastName = driver.findElement(By.id("dtlview_Last Name")).getText();

        if (actLastName.equals(lastName)) {
            System.out.println("Contact Successfully created !!!");
        } else {
            System.out.println("Failed.....");
        }

        //      logout
    }
}
```

```

WebElement profile =
driver.findElement(By.cssSelector("img[src='themes/softed/images/user.PNG']"));

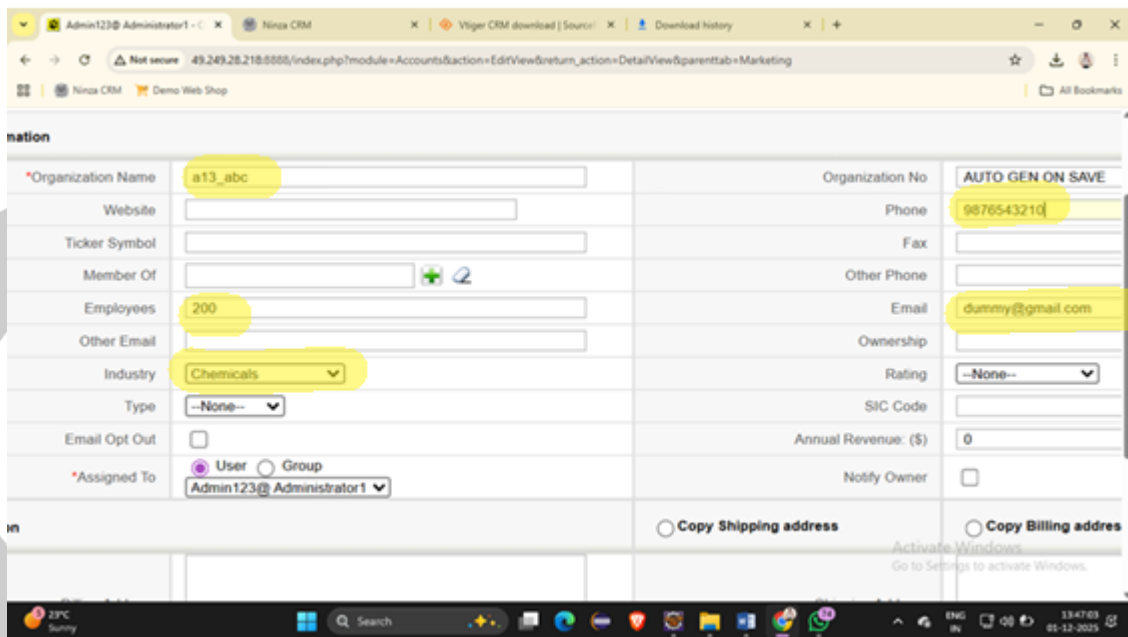
Actions act = new Actions(driver);
act.moveToElement(profile).build().perform();

driver.findElement(By.linkText("Sign Out")).click();

//          close the browser
Thread.sleep(10000);
driver.quit();
}
}

```

5. CreateOrganizationTest.java



```
package organization;
```

```
import java.time.Duration;
```

```

import org.openqa.selenium.By;
import org.openqa.selenium.Keys;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.interactions.Actions;

```

```
import extra.GenRanNum;
```

```

public class CreateOrganizationTest {
    public static void main(String[] args) throws InterruptedException {
//          open the browser
        WebDriver driver = new ChromeDriver();
        driver.manage().window().maximize();
        driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(15));

//          Login

```

```

driver.get("http://localhost:8888/");

WebElement username = driver.findElement(By.name("user_name"));
WebElement password = driver.findElement(By.name("user_password"));
username.sendKeys("admin");
password.sendKeys("password" + Keys.ENTER);

//      Create organization
driver.findElement(By.linkText("Organizations")).click();
driver.findElement(By.cssSelector("img[alt='Create Organization...']")).click();

//      Filling Form
String orgName = "MakunaiGlobal_" + GenRanNum.generateRandomNumber();
WebElement orgField = driver.findElement(By.name("accountname"));
orgField.sendKeys(orgName);

//      Save
driver.findElement(By.cssSelector("input[title='Save [Alt+S]']")).click();

//      Verification
String actOrgName = driver.findElement(By.id("dtlview_Organization Name")).getText();

if (actOrgName.equals(orgName)) {
    System.out.println("Organization Succesfully created !!!");
} else {
    System.out.println("Failed.....");
}

//      logout
WebElement profile =
driver.findElement(By.cssSelector("img[src='themes/softed/images/user.PNG']"));

Actions act = new Actions(driver);
act.moveToElement(profile).build().perform();

driver.findElement(By.linkText("Sign Out")).click();

//      close the browser
Thread.sleep(10000);
driver.quit();
    }
}

```